

What Does a Heart Sound Classifier Actually Learn? A Controlled Comparison of Cepstral, Spectral and Wavelet-Packet Representations with Cardiac-Cycle-Aligned Explanations

Ouardi Ilyass

Dipartimento di Informatica
Università degli Studi di Milano

Milan, Italy

ilyass.ouardi@studenti.unimi.it

Abstract—Automatic screening of phonocardiogram (PCG) recordings could extend cardiac auscultation to settings without expert cardiology, but on the PhysioNet/CinC 2016 corpus a high aggregate score is easy to obtain and hard to trust, because the recording site is itself strongly predictive of the diagnosis. This project compares three feature representations (MFCC, log-Mel spectrogram, perceptual wavelet packet) and three classifiers (SVM, random forest, small CNN) under one recording-level protocol, and pairs each performance claim with a diagnostic that could refute it: unsupervised k -means structure, a per-site breakdown, and a falsifiable test of whether attributions concentrate in systole. The best model reaches a modified accuracy of 0.866, comparable to published entries, but k -means recovers the acquisition site far better than the diagnosis, and per-site modified accuracy ranges from 0.99 down to 0.48. Attribution is only weakly systole-enriched, and no more so for correct than for incorrect predictions. We conclude that the aggregate metric overstates what the models have learned.

Index Terms—phonocardiogram, heart sound classification, explainable AI, Grad-CAM, SHAP, wavelet packet decomposition, MFCC, dataset shortcuts

I. INTRODUCTION

Cardiac auscultation is the oldest and cheapest cardiac screening tool in clinical use, but interpreting heart sounds by ear is difficult and depends heavily on the listener’s training [1], [2]. The two dominant events, the first and second heart sounds (S1 and S2), are short transients carrying most of their energy below 100 Hz; murmurs are noise-like signals occupying the intervals between them and extending to several hundred hertz [3]. Automatic analysis of the phonocardiogram (PCG) is attractive wherever expert cardiology is scarce: primary-care triage, telemedicine, and low-cost digital stethoscopes. The task addressed here is the binary one posed by the PhysioNet/Computing in Cardiology (CinC) 2016 Challenge — label a recording *normal* or *abnormal*, where abnormal denotes a confirmed cardiac pathology rather than merely an audible murmur [4], [5].

Work on this corpus is well developed. The winning entry combined a feature-based classifier ensemble with a convolutional network on band-limited cardiac cycles [6]; a purely

convolutional system on MFCC heat maps followed [7], as did feature ensembles that avoid segmentation altogether [8]. Wavelet-packet descriptions of the PCG form a separate line [9], [10], motivated by the fact that the informative events are transients of tens of milliseconds, for which the local-stationarity assumption behind the short-time Fourier transform fits poorly [11]. Reported modified accuracies cluster around 0.84–0.86.

Less developed is the question of *what* such a system learns. The public portion of CinC 2016 comprises six sub-databases collected by different groups with different equipment, and abnormal prevalence varies across them by roughly 69 pp. Site identity is therefore strongly predictive of the label, and a model that recognises the recording device or the room can score well without learning anything about cardiac acoustics — the failure mode documented for radiographic COVID-19 detectors [12]. Here it is not hypothetical; it is the dominant structure in the data.

This project therefore treats the aggregate metric as a claim to be tested. We implement one recording-level pipeline and use it to compare three representations that differ in kind — cepstral (MFCC), spectral (log-Mel) and perceptual wavelet packet (PWP) — across three classifier families. Each performance claim is paired with a diagnostic that could refute it. k -means asks whether the diagnosis is geometrically present in each feature space *before any label is used*, with cluster agreement measured against both the diagnosis and the recording site so that a null result is still informative. Performance is broken down per sub-database, since a single bright site exposes a shortcut. And the explainability analysis is framed as a falsifiable prediction: because most clinically important adult murmurs are systolic, a detector that has learned pathology should place more attribution mass in systole than uniform temporal attention would, which can be tested against an analytic and a permutation null.

The scope is narrow. We propose no new architecture and make no claim beyond the split we evaluate; the official challenge test set was never released, so all comparison

with published entries is contextual. The contribution is the controlled comparison and the diagnostics — and two of the three expectations we began with were contradicted by the evidence. They are reported here in the same voice as the results that held.

II. METHOD

Fig. 1 shows the pipeline. Its distinguishing property is not any single block but the fact that the three feature domains are computed from *identical* windows and evaluated under a single recording-level protocol, so that any difference between them is attributable to the representation and not to the experimental setup. The remainder of this section describes each stage and the reasoning behind the choices that are not self-evident.

A. Preprocessing

Recordings are resampled to 2 kHz and band-pass filtered to 25–400 Hz with a fourth-order Butterworth design. Below 25 Hz the signal is respiration, movement and baseline drift; above 400 Hz there is essentially no cardiac acoustic energy. The band covers S1 (≈ 25 –45 Hz), S2 (≈ 55 –75 Hz) and murmur energy.

Filtering is applied forward and backward, so that it is zero-phase. This matters specifically for the analysis in Section VI: a causal filter has frequency-dependent group delay, which would shift high-frequency murmur energy relative to the low-frequency S1 fundamental, and the alignment test is precisely a claim about *when* evidence occurs.

Friction transients from the stethoscope diaphragm are suppressed by the spike removal procedure of Schmidt et al. [13]; these are the loudest events in many recordings and would otherwise dominate every energy-based descriptor. Finally each recording is z -normalised. Gain is a site artifact — different clinics used different preamplifiers — and since gain correlates with site and site correlates with prevalence, preserving absolute level would hand the model a shortcut.

B. Windowing

Windows are 3 s long and fixed. At 60–100 bpm a cardiac cycle lasts 0.6–1.0 s, so a 3 s window always contains at least three complete cycles and is robust to the phase at which it happens to start. Training uses 50 % overlap as augmentation; validation and test use non-overlapping windows, so that evaluation samples remain statistically independent and confidence intervals are not artificially narrowed.

Windows are deliberately *not* cut at cardiac-cycle boundaries. Doing so would build the location of systole into the input representation and would make the alignment claim of Section VI circular.

C. Three feature representations

The three representations were chosen to differ in kind rather than in parameter setting.

MFCC. Thirteen cepstral coefficients from a mel filterbank, with first and second time derivatives, each trajectory summarised over the window by six statistics (mean, standard

deviation, minimum, maximum, skewness, kurtosis), giving 234 dimensions. This is the standard baseline in the CinC 2016 literature [7], [14], so it anchors comparability, and the cepstrum is a decorrelated description of spectral shape, which is broadly what separates murmur from no-murmur.

Log-Mel spectrogram. The same filterbank output retained as a two-dimensional time–frequency map and used as CNN input. This is the only representation that preserves the time axis; the statistical pooling used for the classical branch discards it. The contrast between the two is one of the comparisons the study is designed to make.

Perceptual wavelet packet (PWP). A Daubechies-4 wavelet packet decomposition [15] whose uniform subbands are merged into perceptually spaced bands, each described by log energy, relative energy, Shannon entropy, standard deviation, skewness, kurtosis and peak amplitude, following the construction of Safara et al. [9]. MFCC and log-Mel are both STFT-based and assume local stationarity within each window, which fits poorly a signal whose informative events are 10–50 ms transients; the packet transform tiles the time–frequency plane adaptively [10], [11]. A further property matters in Section VI: PWP features map onto frequency bands exactly, with no transform in between.

Two resolution constraints shaped these parameters and are worth stating because they otherwise look arbitrary. First, the number of mel bands is limited by FFT resolution: at 2 kHz with a 256-point FFT the bin width is 7.81 Hz, so only about 48 usable bins fall inside 25–400 Hz, and requesting more filters than that produces empty, all-zero filters that inject structural zeros into every feature vector. Second, wavelet packet nodes are not returned in frequency order: the packet tree reverses the frequency axis in the high-pass branch at each level, so the natural ordering is a bit-reversed permutation of frequency order, and indexing it as though it were frequency-ordered silently scrambles the spectral axis without raising an error. Nodes are reordered explicitly before grouping.

D. Classifiers

Three families with different inductive biases are compared: an RBF-kernel SVM (global, distance-based, maximum-margin); a random forest (local, axis-aligned, ensemble, and additionally admitting an exact SHAP computation [16]); and a small two-dimensional CNN on log-Mel input, comprising four Conv–BN–ReLU–MaxPool blocks, global average pooling and a small fully connected head, on the order of 10^5 parameters. The network is kept deliberately small because the dataset is small; a large pretrained network would memorise it and would obscure the feature-domain comparison that is the point of the study.

Class imbalance is handled by inverse-frequency weighting in all three families. Augmentation uses circular time shift — the phase at which a window starts is arbitrary, so the label must be invariant to it — and additive Gaussian noise. Frequency masking is deliberately avoided: masking a band could delete the murmur itself, converting an abnormal example into a mislabelled normal one.

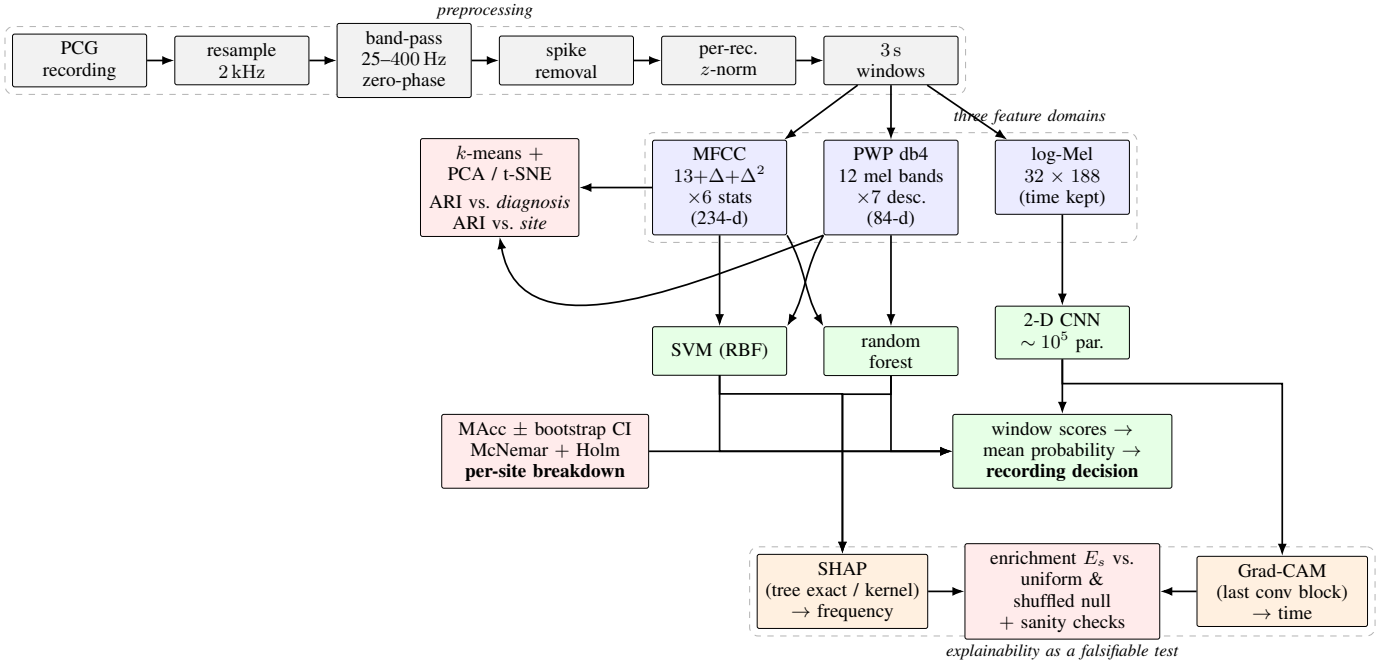


Fig. 1. Pipeline. The three feature domains are extracted from identical windows and evaluated under a single recording-level protocol, so that any difference between them is attributable to the representation. The two shaded blocks on the right are diagnostics rather than outputs: *k*-means asks whether the class structure exists before supervision, and the enrichment test asks whether the CNN’s attribution is placed where cardiac physiology says the evidence should be.

Window scores are aggregated to a recording decision by averaging predicted probabilities, because a clinician diagnoses a patient rather than a three-second excerpt. Majority voting was also computed and gave results within 0.004 MAcc of probability averaging.

III. EXPERIMENTAL SETUP

A. Dataset

We use the public training portion of the PhysioNet/CinC 2016 corpus [4], [17]: 3240 mono recordings, 20.2 h in total, sampled at 2 kHz, with durations from about 5 s to 120 s. Labels are per recording, and 20.5% of recordings are abnormal, an imbalance ratio of 3.87:1. The corpus is divided into six sub-databases, `training-a` through `training-f`, each collected by a different group at a different site with different equipment. Abnormal prevalence differs across them by 68.9 pp, which is the single most important property of this corpus for the present study.

The official challenge test set was never publicly released; it remained on the challenge server. All results below therefore come from a self-defined split of the public training data, and published challenge scores are context rather than a like-for-like comparison. We restate this wherever those numbers appear.

B. Splits

Splits are made at the *recording* level, never at the window level, and stratified jointly on label and sub-database. Recording-level splitting prevents windows of the same patient

— sharing a stethoscope, a noise floor and a murmur — from appearing in more than one partition; this is the most consequential guard in the whole protocol. Joint stratification keeps both the class balance and the site mix constant, so that the test partition cannot end up dominated by one clinic. The realised deviation is at most 0.07 pp in class balance and 0.24 pp in site composition.

Windowing yields 30 695 training, 3329 validation and 3362 test windows, the latter drawn from 484 recordings of which 20.5% are abnormal. No recording was excluded for quality or loading failure.

C. Parameters

Features use a 256-point Hann window with hop 32 (16 ms), 32 mel bands over 25–400 Hz. MFCC uses 13 coefficients with Δ and Δ^2 (width 5) and six aggregation statistics, giving 234 dimensions; log-Mel windows are 32×188 ; PWP uses a level-6 db4 tree grouped into 12 perceptual bands with 7 descriptors each, giving 84 dimensions. All classical features are standardised with statistics fitted on the training split only. No degenerate all-zero mel band and no near-constant feature was produced.

Classical hyperparameters were chosen by five-fold stratified group cross validation on the training split, grouping by recording and scoring balanced accuracy. The selected SVM uses $C = 10$, $\gamma = 5 \times 10^{-4}$, with cross-validated MAcc 0.874 ± 0.008 . The CNN was trained for at most 60 epochs with Adam at 10^{-3} , batch size 64, dropout 0.3, plateau scheduling and early stopping on validation MAcc; it stopped

TABLE I

UNSUPERVISED STRUCTURE OF EACH FEATURE SPACE AT $k = 2$, ON THE TRAINING SPLIT. ARI AGAINST THE DIAGNOSIS IS COMPARED WITH ARI AGAINST THE RECORDING SITE TO IDENTIFY WHAT THE CLUSTERS ACTUALLY TRACK.

Features	Silh.	DB	ARI (class)	ARI (site)	PCA dims (95%)
MFCC	0.396	1.512	0.018	0.201	51
log-Mel	0.192	1.703	0.028	0.043	37
PWP	0.388	1.218	0.002	0.159	31

after 20 epochs with its best epoch at 8 and validation MAcc 0.915. Model selection, hyperparameter search and decision thresholds used training and validation data only; the test partition was read once, after selection was frozen.

D. Metrics and statistics

The primary metric is the official challenge metric, modified accuracy $\text{MAcc} = (\text{Se} + \text{Sp})/2$, which is identical to balanced accuracy. Plain accuracy is reported but is not an appropriate headline: with roughly 80% normal recordings, a model that always predicts *normal* scores 0.795 accuracy at 0.500 MAcc. Trivial baselines appear in Table II so that the reader need not do this arithmetic. Sensitivity, specificity, precision, F1, ROC-AUC and the Matthews correlation coefficient are also reported.

Confidence intervals are obtained by bootstrap over *recordings*, which is the correct unit of independence; resampling windows would treat the several windows of one recording as independent observations and would produce intervals far too narrow. Pairwise model comparison uses McNemar’s test [18], because all models are evaluated on the same recordings and the samples are therefore paired, with Holm–Bonferroni correction across the family of ten comparisons.

IV. RESULTS I: WHAT IS IN THE FEATURE SPACE BEFORE SUPERVISION?

k -means is used here as a diagnostic rather than as a decorative preliminary. It answers a question the classifiers cannot: is the normal/abnormal distinction geometrically present in each feature space *before any label is used*? We sweep $k \in \{2, \dots, 8\}$ on the training split, and measure agreement of the partition against two references — the diagnosis and the recording site — so that a null result against the diagnosis is still informative.

Table I reports the $k = 2$ result. The adjusted Rand index against the diagnosis is 0.018 for MFCC, 0.028 for log-Mel and 0.002 for PWP: all three are close to chance. Against the recording site the same partitions score 0.201, 0.043 and 0.159. For MFCC and PWP the site agreement exceeds the class agreement by an order of magnitude. The same pattern holds at other k : for MFCC at $k = 3$, ARI reaches 0.271 against the diagnosis but 0.420 against the site.

Fig. 2 shows the corresponding PCA and t-SNE projections, in which no domain visibly separates the two classes. We stress that the site/diagnosis asymmetry is established by the index values above and not by these plots: the projections are

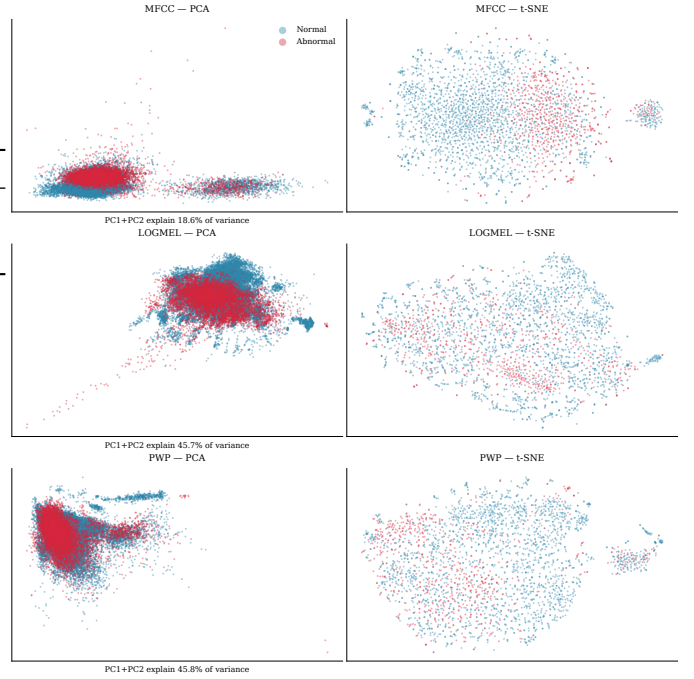


Fig. 2. PCA and t-SNE projections of the three feature spaces, one row per feature domain, coloured by diagnosis. No projection separates the two classes; the first two principal components explain only 18.6% of the variance in the MFCC space.

coloured by diagnosis only, and in any case t-SNE cluster sizes and inter-cluster distances are not quantitatively meaningful, so a projection is at best corroborating evidence for a partition statistic. The comparison of ARI against two references is the actual measurement.

Two observations follow. First, the expectation we started from — that the diagnosis would be recoverable without labels — is contradicted. Second, and more usefully, the null result is not uninformative, because the site comparison identifies what the dominant variance actually encodes. The class boundary in these spaces is a thin supervised direction inside a geometry organised primarily by how and where the recording was made. This is the first of three independent indications pointing the same way, and it is the reason we treat the aggregate metric of Section V with caution rather than reporting it as a headline.

The moderate silhouette values (0.396 for MFCC, 0.388 for PWP) should not be read as evidence of good clustering: at $k = 2$ the MFCC partition is extremely unbalanced (29 089 versus 1606 windows), and a silhouette computed on one large and one small compact cluster is easy to obtain. Davies–Bouldin and Calinski–Harabasz degrade sharply for $k \geq 3$ in every domain, which is consistent with there being no strong natural partition at any k we tested.

V. RESULTS II: CLASSIFICATION

A. Aggregate performance

Table III gives recording-level test performance for the five feature/model pairings, and Table II the trivial baselines that put them in perspective. All five models clearly exceed

TABLE II
TRIVIAL BASELINES ON THE TEST SET (484 RECORDINGS). THE ALWAYS-NORMAL PREDICTOR ILLUSTRATES WHY ACCURACY IS NOT AN APPROPRIATE HEADLINE METRIC HERE.

Baseline	Acc.	Se.	Sp.	MAcc
always normal	0.795	0.000	1.000	0.500
always abnormal	0.205	1.000	0.000	0.500
random stratified	0.638	0.212	0.748	0.480

TABLE III
RECORDING-LEVEL PERFORMANCE ON THE HELD-OUT TEST SET, ORDERED BY MACC. $MAcc = (Se + Sp)/2$ IS THE PRIMARY METRIC; 95 % BOOTSTRAP CONFIDENCE INTERVALS ARE RESAMPLED OVER RECORDINGS. ALL FIVE INTERVALS OVERLAP.

Model	Acc.	Se.	Sp.	MAcc	95 % CI	F1	AUC
SVM-MFCC	0.816	0.949	0.782	0.866	[0.834, 0.894]	0.679	0.922
CNN-log-Mel	0.851	0.889	0.842	0.865	[0.828, 0.900]	0.710	0.961
RF-MFCC	0.802	0.949	0.764	0.857	[0.824, 0.886]	0.662	0.906
SVM-PWP	0.785	0.909	0.753	0.831	[0.794, 0.863]	0.634	0.877
RF-PWP	0.731	0.970	0.670	0.820	[0.790, 0.847]	0.596	0.864

the always-normal predictor: MAcc ranges from 0.820 to 0.866 against 0.500. The Matthews correlation coefficient, which is zero for any constant predictor, is between 0.59 and 0.66 for the leading models. This claim is supported without qualification.

The claim that the CNN would outperform the best classical pairing is not. SVM-MFCC reaches MAcc 0.866 95 % CI [0.834, 0.894] and the CNN 0.865 [0.828, 0.900], a difference of 0.0004 inside intervals that almost entirely coincide. On MAcc the two are statistically indistinguishable, and so, on the interval criterion, are all five models.

The confusion matrices (Fig. 3) show what the aggregate hides: the models differ chiefly in where they sit on the sensitivity/specificity trade-off. RF-PWP achieves the highest sensitivity of all (0.970) at the cost of the lowest specificity (0.670); the CNN is the most balanced (0.889/0.842). For a screening application, where a missed pathology costs more than a referral, the high-sensitivity operating points are arguably preferable, but nothing in our evaluation selects among them. The ROC curves (Fig. 4) add a further nuance: the CNN has the best AUC (0.961 against 0.922 for SVM-MFCC) without the best MAcc, meaning its *ranking* of recordings is better than its chosen operating point. A threshold tuned on validation data captures only part of that advantage.

B. Paired comparison

MAcc intervals and paired error tests do not agree, and the disagreement is informative rather than contradictory. Table IV shows that after Holm-Bonferroni correction the CNN makes significantly fewer recording-level errors than every other model, including SVM-MFCC (28/11 discordant, $p = 1.0 \times 10^{-2}$), while the two MFCC classical models are indistinguishable from each other ($p = 0.21$).

These statements are compatible. McNemar tests the count of *discordant errors* at each model's fixed operating point;

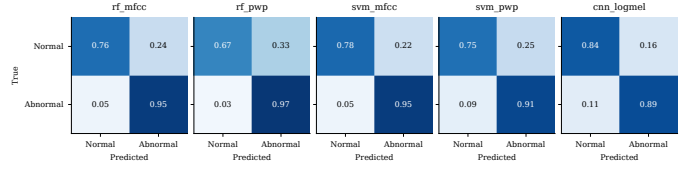


Fig. 3. Recording-level confusion matrices. The models differ mainly in where they sit on the sensitivity/specificity trade-off rather than in overall discrimination.

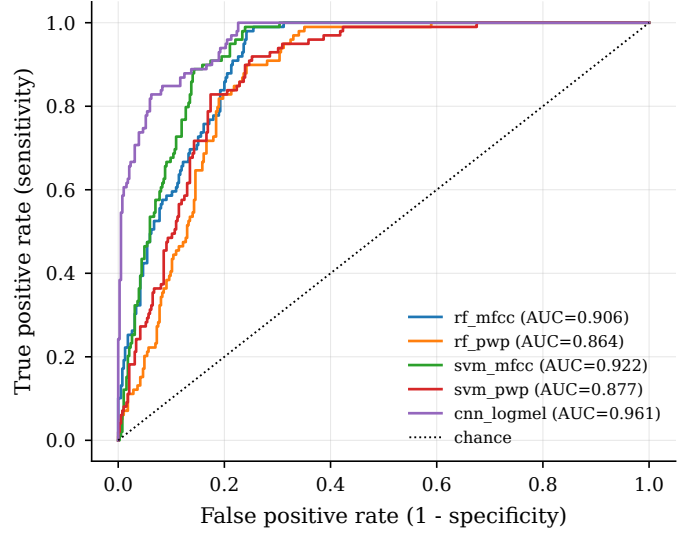


Fig. 4. ROC curves on the test recordings. The CNN attains the highest AUC (0.961) while not attaining the highest MAcc, indicating that its ranking of recordings is better than its operating point.

MAcc averages sensitivity and specificity, which weights the minority class far more heavily. The CNN gets more recordings right overall, but its errors are distributed such that the sensitivity/specificity average is unchanged. Since MAcc is the metric this task is defined by, we do not claim the CNN is the better model; we state that it is better on total errors and equivalent on the primary metric.

The feature domain matters more than the classifier family. Both MFCC models outrank both PWP models, and every MFCC-versus-PWP comparison that is significant favours MFCC. The wavelet-packet representation, adopted on the principled grounds of Section II, did not pay off here. We note in Section VI that it nevertheless remains the more useful representation for interpretation, which is a different property from accuracy.

C. The aggregate metric is not uniform across sites

Table V and Fig. 5 decompose the best model's performance by acquisition sub-database, and this is the result that most changes how the preceding numbers should be read. Performance is not uniform. MAcc is 0.986 on training-e and 0.477 on training-a, a spread of 0.509; on training-a, training-b and training-d the model is at or below the 0.500 of a constant predictor. Specificity collapses to 0.000

TABLE IV

PAIRWISE McNEMAR TESTS ON RECORDING-LEVEL PREDICTIONS. DISCORDANT COUNTS ARE (A CORRECT / B CORRECT). SIGNIFICANCE IS ASSESSED AFTER HOLM–BONFERRONI CORRECTION ACROSS THE FAMILY OF TEN COMPARISONS.

Comparison	Discordant	p	Sig.
CNN–log–Mel vs. RF–PWP	68/10	1.1×10^{-10}	✓
RF–PWP vs. SVM–MFCC	8/49	1.2×10^{-7}	✓
RF–MFCC vs. RF–PWP	39/5	6.5×10^{-7}	✓
CNN–log–Mel vs. SVM–PWP	43/11	2.5×10^{-5}	✓
RF–PWP vs. SVM–PWP	9/35	1.6×10^{-4}	✓
CNN–log–Mel vs. RF–MFCC	35/11	7.0×10^{-4}	✓
CNN–log–Mel vs. SVM–MFCC	28/11	1.0×10^{-2}	✓
SVM–MFCC vs. SVM–PWP	27/12	2.5×10^{-2}	–
RF–MFCC vs. SVM–MFCC	8/15	2.1×10^{-1}	–
RF–MFCC vs. SVM–PWP	24/16	2.7×10^{-1}	–

TABLE V

BEST MODEL (SVM–MFCC) BROKEN DOWN BY ACQUISITION SUB-DATABASE. PERFORMANCE IS NOT UNIFORM: MAcc SPANS 0.509. TRAINING–C ($n = 4$) IS OMITTED AS IT CONTAINS A SINGLE CLASS IN THE TEST SPLIT.

Sub-database	n	Prev.	MAcc	Se.	Sp.
training-a	61	0.721	0.477	0.955	0.000
training-b	73	0.205	0.511	0.867	0.155
training-d	8	0.500	0.500	1.000	0.000
training-e	321	0.087	0.986	1.000	0.973
training-f	17	0.294	0.692	0.800	0.583

on training-a and 0.155 on training-b: on those sites the model labels almost everything abnormal.

The pooled figure is therefore an average dominated by one site. training-e contributes 321 of the 484 test recordings (66%) and has a low abnormal prevalence of 8.7%; it is both the largest and the easiest partition. training-a, by contrast, has 72.1% abnormal prevalence, and a model that answers “abnormal” there is right 72% of the time by accident of prevalence.

This is exactly the behaviour Section IV predicted from the unsupervised geometry, and it is what a site-prevalence shortcut looks like when measured. We are careful about the inference: the per-site breakdown establishes that performance does not transfer across acquisition conditions, which is a measurement. That the mechanism is prevalence-driven site recognition is an inference, supported by the clustering result and by the direction of the failures but not directly demonstrated — we did not, for example, train a site classifier on the same features, which would be the natural confirming experiment and is the clearest gap in this study.

The smaller sub-databases (training-d, $n = 8$; training-f, $n = 17$) carry very little weight, and their figures should be read as tendencies rather than findings. training-c ($n = 4$) contains a single class in the test split and is omitted.

D. Overfitting and calibration

Two weaknesses are worth stating. Both random forests show a large gap between training and cross-validated bal-

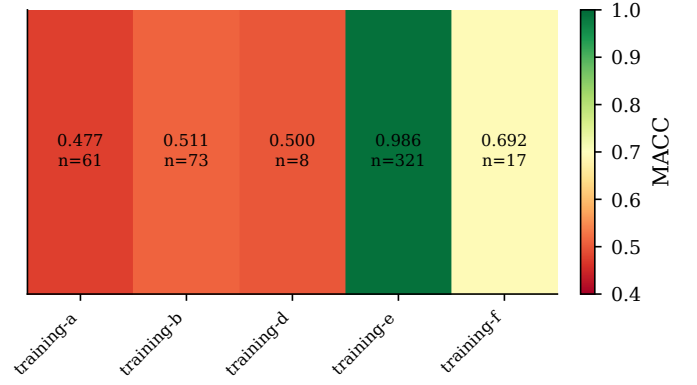


Fig. 5. Per-sub-database MAcc of the best model, with sub-database size. The pooled figure of 0.866 is dominated by training-e, which supplies 66% of the test recordings and on which the model is nearly perfect.

TABLE VI

CONTEXT FROM PUBLISHED PHYSIONET/CINC 2016 ENTRIES. THESE WERE SCORED ON THE OFFICIAL HIDDEN TEST SET, WHICH WAS NEVER RELEASED; OUR NUMBERS COME FROM A HELD-OUT SPLIT OF THE PUBLIC TRAINING DATA. THE COMPARISON IS THEREFORE CONTEXTUAL AND *not* LIKE-FOR-LIKE.

System	MAcc	Se.	Sp.	Same test set
This work (SVM–MFCC)	0.866	0.950	0.782	–
Potes et al. [6]	0.860	0.942	0.778	✓
Zabihi et al. [8]	0.859	0.869	0.849	✓
Rubin et al. [7]	0.839	0.727	0.951	✓

anced accuracy — 0.947 versus 0.867 on MFCC, and 0.945 versus 0.814 on PWP — so they are memorising the training folds despite the depth and leaf-size grid; stronger regularisation would be the obvious next step. Second, the classical models are poorly calibrated, with expected calibration error 0.129 for RF–MFCC and 0.150 for RF–PWP. Since recording-level decisions are formed by averaging window probabilities, miscalibration propagates into the aggregation, and the close agreement between mean-probability and majority-vote aggregation (within 0.004 MAcc) suggests the probabilities are contributing little beyond their ordering.

E. Comparison with published work

Table VI places our best result alongside published CinC 2016 entries. Our MAcc of 0.866 sits at the top of the 0.84–0.86 band those entries occupy. This must not be read as an improvement. The published figures were obtained on the official hidden test set, which was never released; ours come from a self-defined split of the public training data, with a different class balance, a different site mix and a different difficulty. The numbers are of the same order, which is the only conclusion the comparison supports, and it is reassuring mainly as evidence that the pipeline is not broken.

VI. RESULTS III: EXPLAINABILITY AS A TEST

The standard practice of displaying a few heat maps and asserting that a model “focuses on clinically relevant regions” is unfalsifiable as stated. This section states a prediction in

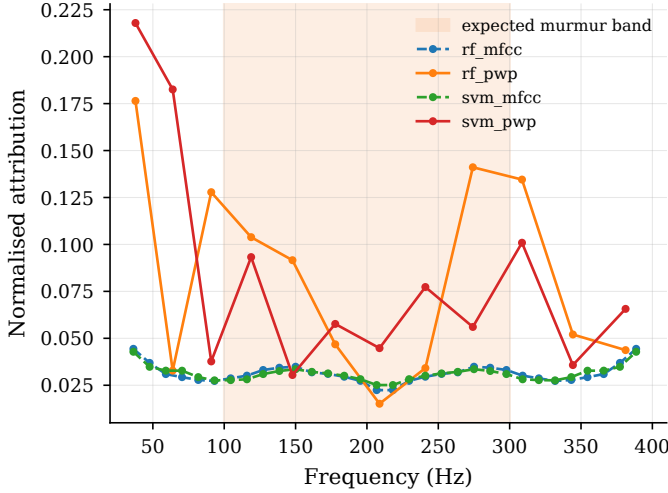


Fig. 6. SHAP attribution mapped to frequency for the four classical feature/model pairings. PWP features map onto bands exactly; the MFCC profile is obtained by redistributing attribution through the magnitude of the DCT basis and is approximate.

advance and then tests it. We use SHAP for the classical models — the exact tree estimator for random forests, the model-agnostic kernel estimator for the SVM [16], [19] — and Grad-CAM at the last convolutional block for the CNN [20].

A. Sanity checks first

Attribution maps mean nothing until they are shown to depend on the trained weights [21], so the checks are reported before the findings. Recomputing Grad-CAM from a randomly re-initialised copy of the network gives maps that correlate with the trained model’s at only 0.051 on average over 100 windows, so the maps reflect learned structure rather than architecture. Correlating the temporal attribution profile with the signal envelope gives 0.068 over 499 windows, with no window above 0.5: attribution is not simply tracking loudness, which matters because S1 and S2 are the loud events and a loudness-driven map would confound any phase-specific claim. Both checks pass.

A third check is less reassuring. Grad-CAM and Grad-CAM++ [22] maps correlate at only 0.392. Our temporal conclusions are therefore specific to the Grad-CAM configuration and should not be assumed to hold for other attribution variants.

B. Where in frequency?

PWP features belong to exactly one frequency band by construction, so their attribution maps onto frequency exactly. MFCC features do not: a cepstral coefficient is a weighted sum over all mel bands, so attribution must be redistributed through the magnitude of the DCT basis. That projection is sign-agnostic and therefore approximate, and where the two disagree the PWP result is the trustworthy one.

All four classical pairings peak at the low edge of the band, 25 Hz, and Fig. 6 shows the profiles. Cross-domain agreement is weak, however: the mean pairwise correlation

TABLE VII
CARDIAC-CYCLE ALIGNMENT OF THE CNN’S GRAD-CAM ATTRIBUTION. E_s IS ENRICHMENT, ATTRIBUTION MASS DIVIDED BY THE TIME FRACTION OCCUPIED BY STATE s ; $E_s = 1$ IS EXACTLY WHAT UNIFORM TEMPORAL ATTENTION PRODUCES. COMPUTED ON THE 66.5 % OF TEST WINDOWS (2235 OF 3362) THAT THE SEGMENTER LABELLED WITH SUFFICIENT CONFIDENCE.

Cardiac state	Attribution mass	E_s
S1	0.190	1.001
systole	0.245	1.056
S2	0.150	0.753
diastole	0.415	1.067

between the four profiles is 0.514, and this average conceals a wide range. The two MFCC models agree almost perfectly with each other (0.961), but RF–PWP and SVM–MFCC agree at only 0.280. The high number is largely a statement about the shared representation rather than about a shared physiological finding: models built on the same features attribute similarly, models built on different features do not. We therefore state agreement across feature domains as weak.

Individually, the leading PWP features are interpretable. The top two by mean absolute SHAP value are the Shannon entropy of band 0 (25–51 Hz, 9.5 % of total attribution) and of band 9 (291–326 Hz, 7.9 %), followed by relative energy in 77–105 Hz. Spectral entropy in a low band and in a high band is a plausible murmur descriptor — murmurs are noise-like where S1 and S2 are tonal. But sign consistency is only 0.57 and 0.55 respectively, meaning these features push toward “abnormal” barely more often than toward “normal”. On that basis we describe them as the features the model relies on most, not as evidence of a murmur detector. For the MFCC models the leading features are MFCC₃ mean and MFCC₀ statistics; the prominence of the zeroth coefficient and its derivative, which describe overall energy and its variation, is not encouraging given Section V-C.

C. Where in the cardiac cycle?

Each window is labelled into S1, systole, S2 and diastole by an envelope-based segmenter that operates independently of the model. Given a unit-mass attribution map, the enrichment for state s is

$$E_s = \frac{\text{fraction of attribution mass inside } s}{\text{fraction of time occupied by } s}, \quad (1)$$

so that $E_s = 1$ is exactly what uniform temporal attention produces, $E_s > 1$ is a genuine preference and $E_s < 1$ is avoidance. The normalisation is the whole point: a statement such as “25 % of the attention fell in systole” is uninterpretable until one knows that systole occupies 23 % of the window. Because most clinically important adult murmurs are systolic, a model that has learned pathology should show $E_{\text{systole}} > 1$ — a prediction made before the measurement.

The prediction is confirmed in sign and is very weak in magnitude. Table VII gives $E_{\text{systole}} = 1.056$, above the uniform null but by about 6 %. Against a permutation null, in which the attribution profile is shuffled in time so that its sparsity and

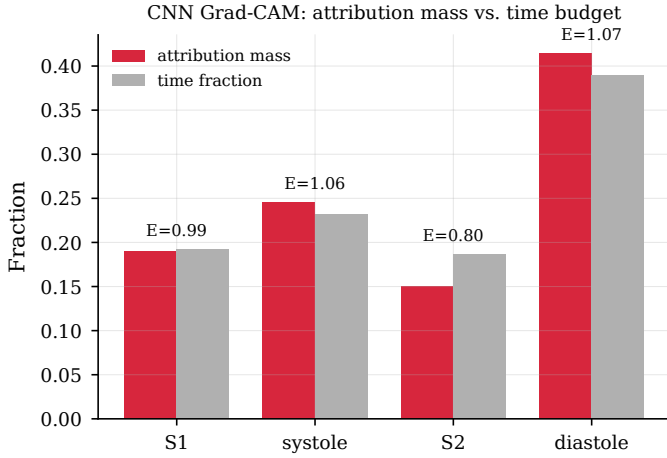


Fig. 7. Distribution of per-window systolic enrichment E_{systole} for the CNN. The dashed line at $E = 1$ is what uniform temporal attention produces. The distribution is shifted slightly right of it.

dynamic range are preserved while any relation to the cardiac cycle is destroyed, only 33% of the 200 tested windows reach significance at $\alpha = 0.05$, the mean p -value is 0.43, and the effect size is Cohen’s $d = 0.131$. Fig. 7 shows the corresponding distribution: shifted right of $E = 1$, but overlapping it heavily. We state this as a tendency, not a property of the model.

Three honesty conditions attach to these numbers. First, the segmenter is a simplified envelope-based labeller, not the reference hidden semi-Markov model [23]. It is used only at evaluation time and never during training, so a noisier segmenter adds variance and makes the test *harder* to pass; it cannot manufacture a positive result. Second, low-confidence windows are excluded, and the honest denominator for every figure above is the 66.5% inclusion rate (2235 of 3362 windows, mean segmenter confidence 0.559, mean estimated heart rate 82 bpm). Third, Grad-CAM’s last convolutional feature map is only a few cells across in each dimension; the frequency axis is too coarse to support spectral claims, so no frequency conclusion is drawn from Grad-CAM. All frequency evidence comes from PWP SHAP.

D. The check that fails

If systolic enrichment reflected murmur detection, it should be stronger when the model is right than when it is wrong. Stratifying by outcome, it is not. Mean E_{systole} is 1.085 on true positives but 1.115 on false positives and 1.157 on false negatives ($n = 26$, so that last figure is a tendency at best). The difference between true and false positives is -0.030 : the model’s systolic focus is, if anything, marginally *stronger* when it raises a false alarm.

This is the clearest negative result in the project, and it constrains what the rest of the section can claim. The attribution maps are not artifacts — the sanity checks establish that — and they do lean toward systole. But that lean does not distinguish correct from incorrect decisions, so it cannot be used as evidence that the model has learned murmur detection.

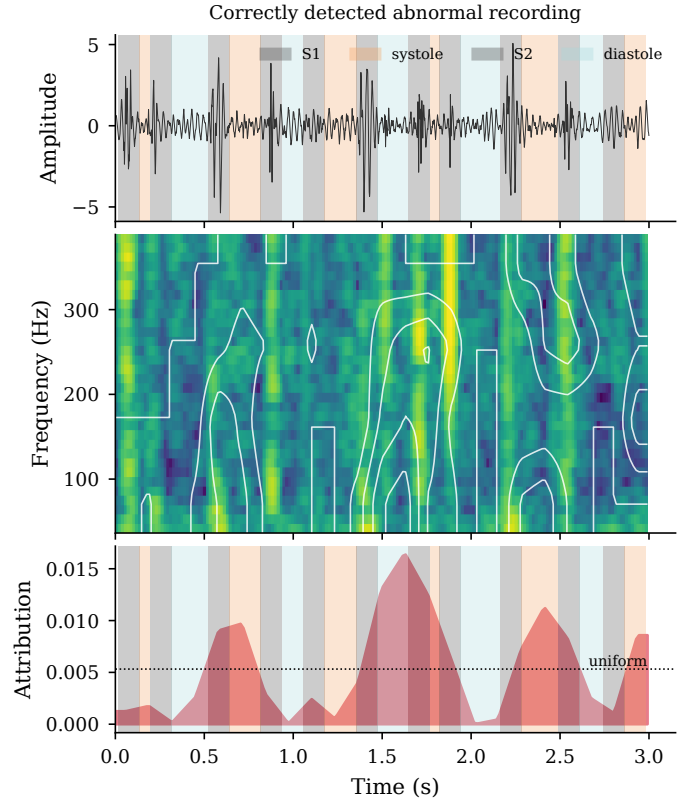


Fig. 8. Grad-CAM on a correctly classified abnormal window, with the segmenter state track below. Maps such as this one motivate the enrichment test; on their own they cannot support it.

Fig. 8 shows a representative correctly classified abnormal window; visual inspection of such maps is what suggested the hypothesis, and the quantitative test is what prevents us from concluding it.

VII. DISCUSSION AND LIMITATIONS

Three diagnostics were applied to a system whose aggregate score, MAcc 0.866, sits comfortably within the published range for this corpus. All three point the same way. Unsupervised clustering recovers the acquisition site far better than the diagnosis in two of three feature spaces. Per-site performance ranges from 0.986 to 0.477, with the pooled figure carried by the single largest and easiest sub-database. And the model’s attribution, while genuinely dependent on learned weights and mildly systole-preferring, is no more systole-preferring on correct decisions than on incorrect ones.

None of these establishes a mechanism individually. Together they make it difficult to maintain that a MAcc near 0.87 on this split demonstrates learned cardiac acoustics, and easy to maintain that acquisition condition is doing substantial work. We state that as an inference, and we name the experiment that would settle it: training a classifier to predict the sub-database from the same features, and measuring how much of the label information survives after site is controlled for. We did not run it.

Several further limitations should be stated plainly rather than waited for. The evaluation rests on one split of one corpus; recording-level splitting and joint stratification protect against the most common leakage, but a single split cannot separate representation effects from split-specific variance, and repeated splits or nested cross-validation would give a firmer footing. Both random forests overfit their training folds, and neither classical model is well calibrated. The MFCC-to-frequency projection used in Section VI is sign-agnostic and approximate, and it disagrees with the exact PWP mapping. Grad-CAM and Grad-CAM++ agree at only 0.392, so the temporal result is specific to one attribution variant. The segmenter labels only 66.5% of test windows with sufficient confidence, and nothing is known about whether the excluded third differs systematically — if low-confidence windows are the noisy ones, they may also be the ones where site cues matter most. Finally, the two smallest sub-databases contribute 8 and 17 test recordings, and their figures carry essentially no statistical weight.

We also note what did work. All five models substantially exceed trivial baselines, and the ranking of feature domains is stable and interpretable: MFCC outperforms PWP under both classifier families, and the log-Mel CNN attains the best ranking quality (AUC 0.961) despite an operating point that gives it no MAcc advantage. The comparison the study was designed to make is therefore answered, even though the answer is not the one anticipated.

VIII. CONCLUSION

This project implemented a heart sound classification pipeline on the public portion of PhysioNet/CinC 2016 and compared three feature representations across three classifier families under a single recording-level protocol. The best configuration, an RBF SVM on MFCC statistics, reaches a modified accuracy of 0.866 on our held-out split, and the log-Mel CNN is statistically indistinguishable from it on that metric while making significantly fewer total recording-level errors.

The more useful findings are the ones that came from testing those numbers rather than reporting them. k -means on the unsupervised feature spaces recovers the recording site better than the diagnosis; per-site modified accuracy spans 0.509, from near-perfect on the largest sub-database to below chance level on two others; and the CNN’s attribution, though verifiably dependent on learned weights and mildly enriched in systole ($E_{\text{systole}} = 1.056$), is not more systole-enriched on correct predictions than on incorrect ones. Two of the three expectations we set out with were contradicted and one was only weakly supported.

We therefore conclude that on this corpus the aggregate metric overstates what the models have learned, and that the diagnostics — not the leaderboard position — are where the informative content of such a study lies. The natural continuation is to make site an explicit variable: predict the sub-database directly from the features to quantify the shortcut, and evaluate under leave-one-site-out rather than stratified

splitting, which measures the generalisation that a screening application would actually require.

REPRODUCIBILITY

The pipeline is driven by a single YAML configuration whose hash is recorded in every result artifact, and every number in this report is read programmatically from those artifacts rather than transcribed by hand. Random seeds are fixed and propagated to NumPy, scikit-learn and PyTorch.

REFERENCES

- [1] M. A. Chizner, “Cardiac auscultation: Rediscovering the lost art,” *Current Problems in Cardiology*, vol. 33, no. 7, pp. 326–408, 2008.
- [2] A. Leatham, *Auscultation of the Heart and Phonocardiography*, 2nd ed. Churchill Livingstone, 1975.
- [3] A. K. Abbas and R. Bassam, “Phonocardiography signal processing,” *Synthesis Lectures on Biomedical Engineering*, vol. 4, no. 1, pp. 1–194, 2009, source for the spectral content of S1, S2 and murmurs.
- [4] C. Liu, D. Springer, Q. Li, B. Moody, R. A. Juan, F. J. Chorro, F. Castells, J. M. Roig, I. Silva, A. E. W. Johnson, Z. Syed, S. E. Schmidt, C. D. Papadaniil, L. Hadjileontiadis, H. Naseri, A. Moukadem, A. Dieterlen, C. Brandt, H. Tang, M. Samieinasab, M. R. Samieinasab, R. Sameni, R. G. Mark, and G. D. Clifford, “An open access database for the evaluation of heart sound algorithms,” *Physiological Measurement*, vol. 37, no. 12, pp. 2181–2213, 2016.
- [5] G. D. Clifford, C. Liu, B. Moody, D. Springer, I. Silva, Q. Li, and R. G. Mark, “Classification of normal/abnormal heart sound recordings: The physionet/computing in cardiology challenge 2016,” *Computing in Cardiology*, vol. 43, pp. 609–612, 2016.
- [6] C. Potes, S. Parvaneh, A. Rahman, and B. Conroy, “Ensemble of feature-based and deep learning-based classifiers for detection of abnormal heart sounds,” in *Computing in Cardiology Conference (CinC)*, 2016, pp. 621–624, challenge winner, MAcc 0.860.
- [7] J. Rubin, R. Abreu, A. Ganguli, S. Nelaturi, I. Matei, and K. Sricharan, “Classifying heart sound recordings using deep convolutional neural networks and mel-frequency cepstral coefficients,” in *Computing in Cardiology Conference (CinC)*, 2016, pp. 813–816, mAcc 0.839.
- [8] M. Zabihi, A. B. Rad, S. Kiranyaz, M. Gabbouj, and A. K. Katsaggelos, “Heart sound anomaly and quality detection using ensemble of neural networks without segmentation,” in *Computing in Cardiology Conference (CinC)*, 2016, pp. 613–616, mAcc 0.859; notable for avoiding segmentation entirely.
- [9] F. Safara, S. Doraisamy, A. Azman, A. Jantan, and A. R. Abdullah Ramaiah, “Multi-level basis selection of wavelet packet decomposition tree for heart sound classification,” *Computers in Biology and Medicine*, vol. 43, no. 10, pp. 1407–1414, 2013, perceptual wavelet packet features for PCG.
- [10] R. R. Coifman and M. V. Wickerhauser, “Entropy-based algorithms for best basis selection,” *IEEE Transactions on Information Theory*, vol. 38, no. 2, pp. 713–718, 1992.
- [11] S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way*, 3rd ed. Academic Press, 2008.
- [12] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, “Ai for radiographic covid-19 detection selects shortcuts over signal,” *Nature Machine Intelligence*, vol. 3, no. 7, pp. 610–619, 2021, motivates our per-site shortcut analysis.
- [13] S. E. Schmidt, C. Holst-Hansen, C. Graff, E. Toft, and J. J. Struijk, “Segmentation of heart sound recordings by a duration-dependent hidden markov model,” *Physiological Measurement*, vol. 31, no. 4, pp. 513–529, 2010, source of the spike-removal procedure used in preprocessing.
- [14] S. Davis and P. Mermelstein, “Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980, original MFCC formulation.
- [15] I. Daubechies, “Orthonormal bases of compactly supported wavelets,” *Communications on Pure and Applied Mathematics*, vol. 41, no. 7, pp. 909–996, 1988.
- [16] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, “From local explanations to global understanding with explainable ai for trees,” *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020, treeSHAP.

- [17] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [18] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998, justifies McNemar's test for paired classifier comparison.
- [19] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [20] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.
- [21] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, model-randomisation test used in our phase 08.
- [22] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018, pp. 839–847.
- [23] D. B. Springer, L. Tarassenko, and G. D. Clifford, "Logistic regression-hsmm-based heart sound segmentation," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 4, pp. 822–832, 2016.